

To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

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[varnikakomali@VARNIKAs-MacBook-Air Skill % nano skill_8.c
[varnikakomali@VARNIKAs-MacBook-Air Skill % gcc skill_8.c -o skill_8 && ./skill_8
=== Part 1: Variable Reference Detection and Expansion ===

Input:    Hello $ROLE welcome to $PROJECT
Expanded: Hello Developer welcome to OS_Lab_Practical

Input:    Target config: $UNDEFINED_KEY/backup
Expanded: Target config: /backup

Input:    Path: /home/$USER/workspace
Expanded: Path: /home/varnikakomali/workspace

Input:    $PROJECT/$ROLE/build
Expanded: OS_Lab_Practical/Developer/build

=== Part 2: Built-in Command Dispatch Table Execution ===

Executing 'pwd':
[Built-in pwd] Current directory: /Users/varnikakomali/OS_LAB/skill

Executing 'export LAB_NUM=8':
[Built-in export] Environment variable set: LAB_NUM=8

Executing 'help':
[Built-in help] Supported built-ins: cd, pwd, export, help, exit

Executing 'cd ..':
[Built-in cd] Directory changed to: /Users/varnikakomali/OS_LAB

Executing 'invalid_builtin':
[Dispatch Status] Command not found in built-in dispatch table.
varnikakomali@VARNIKAs-MacBook-Air Skill %
```